

Omar Sharif

PRESIDENTIAL GRADUATE FELLOW | DEPARTMENT OF COMPUTER SCIENCE | DARTMOUTH COLLEGE

✉ omar.sharif.gr@dartmouth.edu | 🏠 omar-sharif.github.io | 🌐 omar-sharif | in omar1303



Education

Sep 2022 - May 2027	Ph.D. in Computer Science Specialization: Natural Language Processing Advisor: Nikhil Singh 🔗 Guarini School of Graduate and Advanced Studies, Dartmouth College
Aug 2019 - Jan 2022	Masters in Computer Science & Engineering Thesis: Fine-grained Categorization of Aggressive Texts using Transformers Advisor: Md. Moshikul Hoque 🔗 Chittagong University of Engineering and Technology (CUET), Bangladesh
Mar 2014- Nov 2018	Bachelors in Computer Science & Engineering CGPA: 3.90/4.0 Chittagong University of Engineering and Technology (CUET), Bangladesh



Research Experience

Mar 2025 - June 2025	Research Intern, Amazon AWS a > Worked on improving the visual reasoning abilities of multimodal models using RL Mentors: Alexandar Li, Sijia Wang, Patrick Ng 🔗
Sep 2022 - Ongoing	Graduate Research Assistant , Department of Computer Science, Dartmouth College
Jul 2022 - Aug 2022	Assistant Professor , Department of Computer Science & Engineering, CUET
Jul 2019 - Jun 2022	Lecturer , Department of Computer Science & Engineering, CUET
Dec 2018 - Jun 2019	Research Assistant , CUET NLP Lab, Bangladesh



Selected Publications

Note: This list includes papers published in peer-reviewed venues across computer science (e.g., AACL, ACL, EMNLP, NAACL). These venues are highly selective and comparable in prestige to top journals in other fields.

To get the full list of please check: [G \[Google Scholar\]](#) [🎓 \[Semantic Scholar\]](#)

(*) indicates equal contribution.

Pre-Prints, Under Review & In-Preparation

- P3. Improving Information Extraction with Learned Queries
Omar Sharif, Soroush Vosoughi, Nikhil Singh
- P2. From Sight to Insight: Improving Visual Reasoning Capabilities of Multimodal Models via Reinforcement Learning
Omar Sharif, Eftekhari Hossain, Patrick Ng
Pre-Print [🔗](#)
- P1. Socially Constructed Treatment Plans: Analyzing Online Peer Interactions to Understand How Patients Navigate Complex Medical Conditions
Madhusudan Basak, **Omar Sharif**, Jessica Hulsey, Elizabeth Saunders, Daisy Goodman, Luke J. Archibald, Sarah Preum
Pre-Print [🔗](#)

Peer Reviewed Conference Papers

- C12. REGen: A Reliable Evaluation Framework for Generative Event Argument Extraction [🔗](#)
Omar Sharif, Joseph Gatto, Madhusudan Basak, Sarah Preum
EMNLP-2025: Findings of the 2025 Conference on Empirical Methods in Natural Language Processing
- C11. Document-Level Event-Argument Data Augmentation for Challenging Role Types [🔗](#)
Joseph Gatto, **Omar Sharif**, Parker Seegmiller, Sarah Preum
ACL-2025: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics
- C10. BenNumEval: A Benchmark to Assess LLMs' Numerical Reasoning Capabilities in Bengali [🔗](#)
Kawsar Ahmed, Md Osama, **Omar Sharif**, Eftekar Hossain, Mohammed Moshikul Hoque
ACL-2025: Findings of the 63rd Annual Meeting of the Association for Computational Linguistics
- C9. Explicit, Implicit, and Scattered: Revisiting Event Extraction to Capture Complex Arguments [🔗](#)
Omar Sharif, Joseph Gatto, Madhusudan Basak, Sarah Preum
EMNLP-2024: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing
- C8. Characterizing Information Seeking Events in Health-Related Social Discourse [🔗](#) [[🔗 Media Coverage](#)]
Omar Sharif, Madhusudan Basak, Tanzia Parvin, Ava Scharfstein, Alphonso Bradham, Jacob T. Borodovsky, Sarah E. Lord, Sarah Preum
AAAI-2024: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence
- C7. Deciphering Hate: Identifying Hateful Memes and Their Targets [🔗](#)
Eftekar Hossain, **Omar Sharif**, Mohammed Moshikul Hoque, Sarah Preum
ACL-2024: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics
- C6. Theme-driven Keyphrase Extraction to Analyze Social Media Discourse [🔗](#)
Omar Sharif*, William Romano*, Madhusudan Basak, Joseph Gatto, Sarah Preum
ICWSM-2024: Proceedings of the Eighteenth International AAAI Conference on Web and Social Media
- C5. A Multimodal Framework to Detect Target Aware Aggression in Memes [🔗](#)
Shawly Ahsan, Eftekar Hossain, **Omar Sharif**, Avishek Das, Mohammed Moshikul Hoque, M. Dewan
EACL-2024: Proceedings of the 18th Conference of the European Chapter of the ACL
- C4. Align before Attend: Aligning Visual and Textual Features for Multimodal Hateful Content Detection [🔗](#)
Omar Sharif*, Eftekar Hossain*, Mohammed Moshikul Hoque, Sarah Preum
EACL-SRW 2024: Proceedings of the 18th Conference of the European Chapter of the ACL: Student Research Workshop
- C3. Chain-of-Thought Embeddings for Stance Detection on Social Media [🔗](#)
Joseph Gatto, **Omar Sharif**, Sarah Preum
EMNLP-2023: Findings of the 2023 Conference on Empirical Methods in Natural Language Processing
- C2. MemoSen: A Multimodal Dataset for Sentiment Analysis of Memes [🔗](#)
Eftekar Hossain, **Omar Sharif**, Mohammed Moshikul Hoque
LREC-2022: Proceedings of the 13th Conference on Language Resources and Evaluation
- C1. Align and Conquer: An Ensemble Approach to Classify Aggressive Texts from Social Media [🔗](#)
Omar Sharif, Mohammed Moshikul Hoque
SPICSCON-2021: IEEE International Conference on Signal Processing, Information, Communication & Systems

Journals Articles

- J5. Tackling Cyber-Aggression: Identification and Fine-Grained Categorization of Aggressive Texts on Social Media using Weighted Ensemble of Transformers [🔗](#)
Omar Sharif, Mohammed Moshikul Hoque
Neurocomputing [Impact Factor: 6.50]

- J4. Information Needs for Opioid Use Disorder Treatment Using Buprenorphine Product: Qualitative Analysis of Suboxone-Focused Reddit Data [🔗](#)
 Madhusudan Basak, **Omar Sharif**, Sarah E Lord, Jacob T Borodovsky, Lisa A Marsch, Sandra A Springer, Edward V Nunes, Charles D Brackett, Luke J Archibald, Sarah Preum
JMIR [Impact Factor: 6.00]: Journal of Medical Internet Research
- J3. Identification of Multilingual Offense and Troll from Social Media Memes using Weighted Ensemble of Multimodal Features [🔗](#)
 Eftekhari Hossain, **Omar Sharif**, Mohammed Moshikul Hoque, M. Ali Akber Dewan, Nazmul Siddique, Md. Azad Hossain
JKSU [Impact Factor: 8.83]: Journal of King Saud University - Computer and Information Sciences
- J2. Multi-class Textual Emotion Categorization using Ensemble of Convolutional and Recurrent Neural Network [🔗](#)
 Tanzia Parvin, **Omar Sharif**, Mohammed Moshikul Hoque
SN Computer Science [Impact Factor: 2.19]
- J1. BEmoC: A Corpus for Identifying Emotion in Bengali Texts [🔗](#)
 MD. Asif Iqbal, Avishek Das, **Omar Sharif**, Mohammed Moshikul Hoque, Iqbal H. Sarker
SN Computer Science [Impact Factor: 2.19]

Workshops

- W7. Identification and Classification of Textual Aggression in Social Media: Resource Creation and Evaluation [🔗](#)
Omar Sharif, Mohammed Moshikul Hoque [[★ Best Paper Award](#)]
CONSTRAINT@AAAI-2021: Combating Online Hostile Posts in Regional Languages during Emergency Situation
- W6. Text Encoders Lack Knowledge: Leveraging Generative LLMs for Domain-Specific Semantic Textual Similarity [🔗](#)
 Joseph Gatto, **Omar Sharif**, Parker Seegmiller, Philip Bohlman, Sarah Preum
GEM@EMNLP-2023: Proceedings of the Third Workshop on Natural Language Generation, Evaluation, and Metrics
- W5. Do LLMs Find Human Answers To Fact-Driven Questions Perplexing? A Case Study on Reddit [🔗](#)
 Parker Seegmiller, Joseph Gatto, **Omar Sharif**, Madhusudan Basak, Sarah Preum
Real-Info@ICWSM-2024: Proceedings of the 1st Workshop on Reliable Evaluation of LLMs for Factual Information
- W4. MUTE: A Multimodal Dataset for Detecting Hateful Memes [🔗](#)
 Eftekhari Hossain, **Omar Sharif**, Mohammed Moshikul Hoque
AAACL-SRW 2022: Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the ACL: Student Research Workshop
- W3. Emotion Classification in a Resource Constrained Language Using Transformer-based Approach [🔗](#)
 Avishek Das, **Omar Sharif**, Mohammed Moshikul Hoque, Iqbal H. Sarker
NAACL-SRW 2021: 2021 Conference of North American Chapter of the ACL: Student Research Workshop
- W2. Offensive Language Detection from Multilingual Code-Mixed Text using Transformers [🔗](#)
Omar Sharif, Eftekhari Hossain, Mohammed Moshikul Hoque
DravidianLangTech@EACL-2021: First Workshop on Speech and Language Technologies for Dravidian Languages
- W1. Multilingual Code-Mixed Hope Speech Detection using Cross-lingual Representation Learner [🔗](#)
 Eftekhari Hossain, **Omar Sharif**, Mohammed Moshikul Hoque [[🏆 Top Model in the Challenge](#)]
LTEDI@EACL-2021: Proceedings of the First Workshop on Language Technology for Equality, Diversity and Inclusion

Awards, Grants, & Honors

2022	Presidential Graduate Fellow: Awarded to only two incoming graduate students at Dartmouth.
2021	Best Paper Award: CONSTRAINT workshop collocated with AAAI-2021.
2022	Best Research Publication Award: Outstanding research paper in the ECE department at CUET in 2022.
2025	Outstanding TA Award: For teaching contribution in the CS Department at Dartmouth.
2024	AAAI Travel Grant: To present our paper in ICWSM-24.
2024	Guarini Travel Grant: To present paper at EMNLP-24.
2024	OpenAI Researcher Access Grant: \$5,000 API credit to assess OpenAI models on human-centric NLP tasks.
2020-2021	Co-PI, ICT division Bangladesh research grant (BDT 10,00,000), > “Detection and Classification of Aggressive Bengali Text on Social Media”
2019-2020	Co-PI, CUET IT business incubator research grant (BDT 4,00,000) > “Automatic Detection of Suspicious Bangla Text using Machine Learning Techniques”
2015-2017	CUET Ex-students merit award: For obtaining the highest GPA in an academic year
2021-2024	Volunteer or D&I Award (ICWSM-24, EACL-[24, 21], EMNLP-[23, 21], ACL-22): To attend and present our papers
2021	Winner: Hope speech detection shared task, EACL-2021
2021	2nd runner-up: Multimodal meme detection and offensive language classification shared task, EACL-2021
2017	27th position (among 1600+ teams): ACM ICPC Dhaka Regional Preliminary
2017	31st position (among 150 selected teams): National Collegiate Programming Contest
2017	Outstanding internship trainee award: For doing the best internship team project

Teaching

I received **Outstanding TA Award (2025)**, for my teaching contributions in the CS Department at Dartmouth.

Spring 26	Head Teaching Assistant , CS 70: Foundations of Applied Computer Science > Class Size: 50, Course 	<i>Dartmouth</i>
Winter 26	Graduate Teaching Assistant , CS 10: Problem Solving via Object Oriented Programming > Class Size: 76, Course 	<i>Dartmouth</i>
Fall 23 & 24	Head Teaching Assistant , CS 74/274: Machine Learning and Statistical Data Analysis > Class Size: 150, Course 	<i>Dartmouth</i>
Winter 23	Graduate Teaching Assistant , CS 39: Theory of Computation > Class Size: 20, Course 	<i>Dartmouth</i>
Fall 22	Graduate Teaching Assistant , CS 30: Algorithms > Class Size: 55, Course 	<i>Dartmouth</i>
Fall 20 & 21	Co-instructor , CS 143: Object-oriented programming (C++) > Class Size: 132, Course 	<i>CUET</i>
Fall 20 & 21	Co-instructor , CS 144: Object-oriented programming (lab) > Class Size: 132, Course 	<i>CUET</i>
Spring 20 & 21	Co-instructor , CS 141: Structured programming (C) > Class Size: 130, Course  ; Lab 	<i>CUET</i>
Summer 21	Co-instructor , CS 326: Internet programming (lab) > Class Size: 65, Course 	<i>CUET</i>

Mentoring & Leadership

Mentoring

I have mentored 15+ undergraduates from CUET and Dartmouth. Collectively, they have co-authored 10+ publications and several now in U.S. PhD programs. Notable mentees include:

CUET	Avishek Das – PhD Student, Kansas State University	[J1, W3]
	Adrita Barua – PhD Student, Kansas State University	
	Fatima Jahara – PhD Student, Rutgers University	[J1]
	Md Asif Iqbal – AL & ML Consultant, Workera	[J1]
	Jawad Hossain – PhD Student, University of Albany	
	Kawsar Ahmed – RA, CUET NLP Lab	[C10]
	Shawly Ahsan – RA, CUET NLP Lab	[C5]
Dartmouth	Ahasan Kabir – PhD Student, University of Central Florida	
	Ava Scharfstein – Data Scientist, JHU Applied Physics Lab	[C8]
	Alphonso Bradham – SWE, Google	[C8]
	Reet Kothari – Data Scientist @ Dozee	

Leadership

Jan 2019 - Aug 2022	Coach , Competitive Programming Club, CUET - Teach students advance data structures, number theory and algorithms
Jan 2019 - Aug 2022	Vice Chair , ACM Student Chapter, CUET - Lead a team of 30+ active members. Plan and asses the growth of the chapter - Arrange research seminars, talks and workshops on different trending topics
Aug 2019 - Aug 2022	Mentor , CUET NLP Lab - Meet weekly with the undergrads working under CUET NLP lab. Discuss research problems, share ideas and track their progress

Technical/Programming Skills

Languages	C, C++, Python, PyTorch
Libraries	Huggingface, LangCahin, vLLM, Tinker, NLTK, Scikit-learn, Gensim
LLM Skills	Supervised Fine-Tuning, Instruction Fine-Tuning, Parameter Efficient Fine-Tuning, QLora, Retrieval Augmented Generation
Research Areas	LLM Evaluation, Information Extraction, Discourse Understanding, Semantic Textual Similarity, Representation Learning, Multimodal Learning, Visual Reasoning, NLP for Social Good

Solved 1000+ competitive programming problems in different online platforms as part of my preparation for proramming contests (ICPC, NCPC, etc). [🔗 Solved problems \(GitHub\)](#)

Codeforces	Solved 500+ problems	UVA	Solved 200+ problems
LightOJ	Solved 120+ problems	CS Academy	Solved 50 problems

Misc

Reviewing	<i>Conferences:</i> EMNLP [23-25], ACL[23-26], ICWSM [23, 24], AAAI-24, LREC [22, 24], NAACL [24-25] <i>Journals:</i> TALLIP, IEEE Access, Scientific Reports, TCSS, LREV, ACM Computing Surveys
Organization	Workshop on Reliable Evaluation of LLMs for Factual Information, ICWSM-24 CUET National Collegiate Programming Contest 2017 CUET CSE Fest 2017, 2018, 2019